

Gestational Diabetes Mellitus



Introduction

Impaired Glucose Tolerance (IGT) with onset or first recognition during pregnancy.

Out of 100 pregnant women in different parts of the country, 10-14 women have GDM

Prevalence of GDM in Indian women is high

Gol recommends screening of all women during ANC, this is part of essential antenatal care package

Effects of GDM on Pregnant Woman and Newborn

Effect on Mother	Effect on Newborn
-Spontaneous abortion	-Intra-uterine death
-Polyhydramnios leading to preterm or premature rupture of membranes	-Fetal macrosomia
-Pre-eclampsia, fits	-Shoulder dystocia
-Prolonged labor leading to assisted delivery	-Birth Injuries, stillbirth
-Obstructed labor leading to caesarean section	-Infant respiratory distress syndrome
-Uterine atony leading to PPH	-Neonatal hypoglycemia and hypocalcemia
-Infection	- Congenital malformation

Oral Glucose Tolerance Test (OGTT)

1 Pregnant Woman



2 Items Required

S.S. tray containing:

- 75 g glucose packet
- 500 ml measuring flask (plastic) with 300 ml water
- Glass
- Spoon



3 Mix Glucose with Water

Dissolve the entire 75 g glucose in 300 ml of water and stir well.



4 Drink the Solution

The pregnant woman should drink the complete glucose solution.



5 Wait for 2 Hours

No food or drink during this 2 hour waiting period (only water is allowed).



6 Check Blood Sugar After 2 Hours

Check blood sugar 2 hours after drinking the solution.



Consideration for GDM Screening

- Single step testing using 75 gm oral glucose & measuring blood sugar 2 hours after ingestion.
- 75 gm glucose is to be given orally after dissolving in approximately 300 ml water **whether the pregnant women comes in fasting or non-fasting state, irrespective of the last meal.**
- The intake of the solution has to be completed within 5-10 minutes.
- If vomiting occurs within 30 minutes of oral glucose intake, the test has to be repeated the next day, or else refer to a facility. If vomiting occurs after 30 minutes, the test continues.

Universal Testing of all pregnant women for GDM

Pregnant Women in Community

Testing for GDM at 1st Antenatal contact (75 gm oral glucose in 300 ml water, 2 hrs later blood sugar (BS) level)

Positive, 2 hrs BS 140 mg/dL or more

Manage as GDM as per GOI guidelines

Negative, 2 hrs BS less than 140 mg/dL

Repeat Test at 24-28 weeks

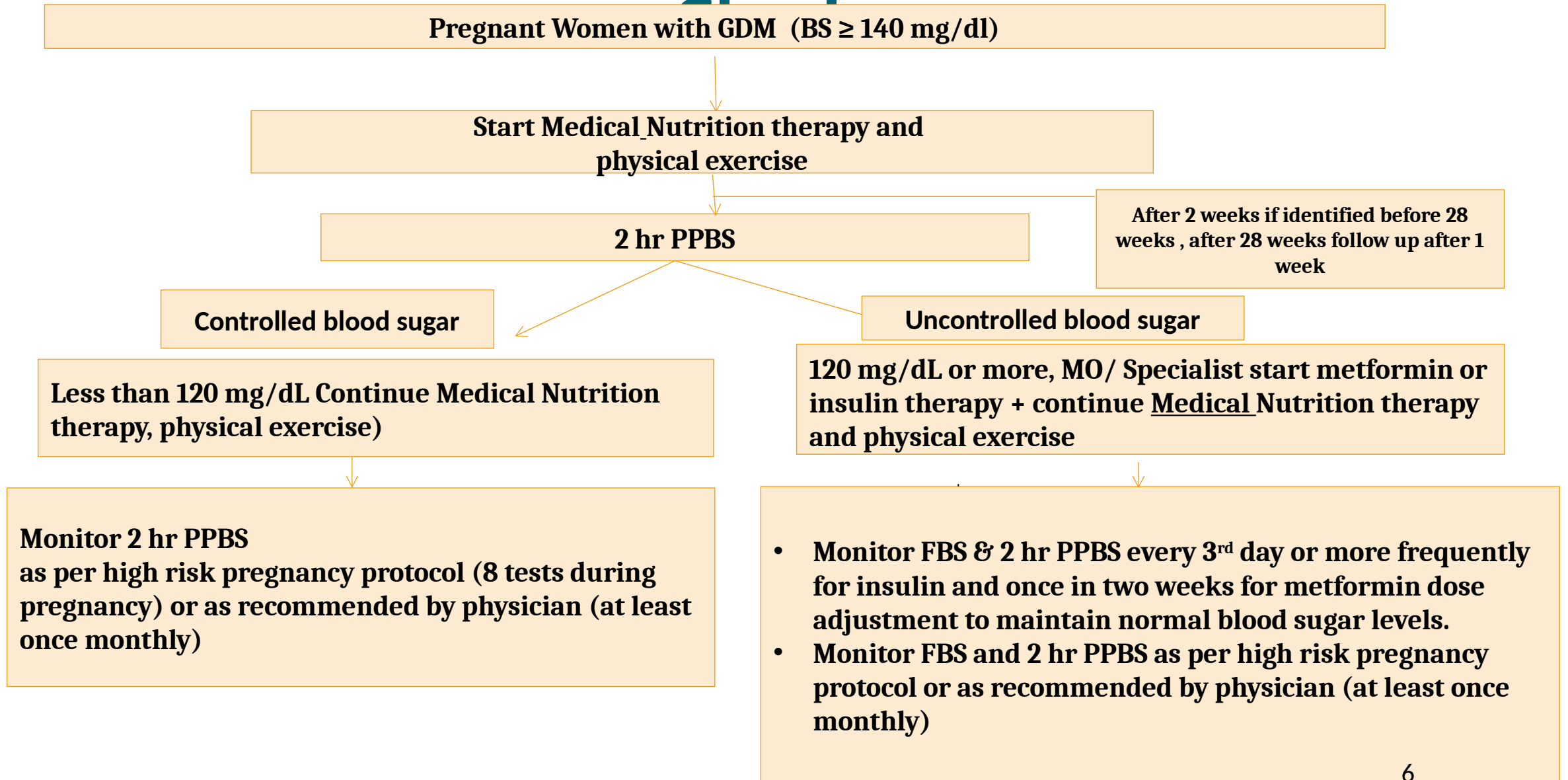
Positive, 2 hrs BS 140 mg/dL or more

Manage as HRP ANC

Negative, 2 hrs BS less than 140 mg/dL

Manage as normal ANC

Management of GDM Flow



Choice of drugs for GDM management

Aim should be to maintain
FBS less than 95 mg/dl and
PPBS less than 120 mg/dl

At CHC/DH/MC, a
Specialist/Gynaecologist/
Physician/MO can start
metformin or insulin.

1. Hypoglycaemia and weight gain with metformin are less in comparison to Insulin.
2. If Insulin is required in high doses, metformin may be added to the treatment.

Insulin is the first drug of choice any time during pregnancy and Metformin (stat 500mg twice daily, maximum upto 2 gm) can be considered after 20 weeks of gestation for medical management of GDM.

Recording

MCP Card

ଖ.୨	ବର୍ତ୍ତମାନର ଗର୍ଭାବସ୍ଥା	
ଖ.୨.୧	ଏହି ଗର୍ଭବତୀର ବୟସ ୧୮ ବର୍ଷରୁ କମ୍ ବା ୩୫ ବର୍ଷରୁ ଉର୍ଦ୍ଧ୍ୱ କି ?	ହଁ / ନା
ଖ.୨.୨	ଆଲୋସିମିଆ / ରକ୍ତ ଶିକୁଳି ରୋଗ ଅଛି କି ?	ହଁ / ନା
ଖ.୨.୩	ଆଇରୀସ୍ ରୋଗ ଅଛି କି ?	ହଁ / ନା
ଖ.୨.୪	ଗର୍ଭବତୀଙ୍କର Rh -ve କି ?	ହଁ / ନା
ଖ.୨.୫	ଉଚ୍ଚତା ୧୪୦ ସେ.ମି. ରୁ କମ୍ ଅଛି କି ?	ହଁ / ନା
ଖ.୨.୬	ଓଜନ ୪୦ କେ.ଜି.ରୁ କମ୍ ?	ହଁ / ନା
ପ୍ରତ୍ୟେକ ଗର୍ଭବତୀଙ୍କର ଏ, ଏନ୍.ସି ସମୟରେ ବିପଦ ସଂକୂଳ ଅବସ୍ଥାର ଆକଳନ		
ଖ.୨.୭	ଗୁରୁତର ରକ୍ତହୀନତା (Hb<7g/dl)	
ଖ.୨.୮	ଗର୍ଭାବସ୍ଥାରେ ଉଚ୍ଚ ରକ୍ତଚାପ (B.P > 140/90) ଅଛି କି ?	
ଖ.୨.୯	ପରିସ୍ରାରେ ଶର୍କରା / ଆଲ୍‌ବୁମିନ୍ ଅଛି କି ?	
ଖ.୨.୧୦	ଗର୍ଭାବସ୍ଥାରେ ମ୍ୟାଲେରିଆ / ଜ୍ୱର ଅଛି କି ?	
ଖ.୨.୧୧	ପ୍ରଥମ ଟ୍ରାଇମେଷ୍ଟର ଠାରୁ ମାସିକ ଓଜନ ୫୦୦ ଗ୍ରାମରୁ କମ୍ ୩ କେଜିରୁ ଅଧିକ ବଢ଼ିଛି କି ?	
ଖ.୨.୧୨	ହେପାଟାଇଟିସ୍ ବି / ଜଣ୍ଡିସ୍ ଅଛି କି ?	
ଖ.୨.୧୩	ଗର୍ଭାବସ୍ଥାରେ HIV+ / RTI / STI ଅଛି କି ?	
ଖ.୨.୧୪	ଯୋନିରୁ ରକ୍ତାବସ୍ଥା ହେଉଛି କି ?	

ଗର୍ଭାବସ୍ଥା ରେକର୍ଡିଂ କାର୍ଡ

ପ୍ରତିଥର ଉଚ୍ଚ ଏବଂ ପରିସ୍ରା ପରୀକ୍ଷା କରାଇ ନିଅନ୍ତୁ । ଯଦି କୌଣସି ଅସ୍ୱାଭାବିକ ପ୍ରକାଶିତ ହେଉଥିବା ବେଳେ ତୁରନ୍ତ ନିକଟସ୍ଥ ସ୍ୱାସ୍ଥ୍ୟକେନ୍ଦ୍ରକୁ ପ୍ରେରିତ କରନ୍ତୁ ।

ନେମୋଗ୍ରାଫିକ୍
ପରିସ୍ରାରେ ସର୍କରା ଓ
ଆଲ୍‌ବୁମିନ୍ ଏବଂ ASB
ଆଲ୍‌ବୁମିନ୍
(ମ୍ୟାଲେରିଆ ପ୍ରଦହନ ଅଞ୍ଚଳ ପାଇଁ)
ବଡ଼ ଗୁପ୍ତ ଏବଂ ଆଇ.ଏଚ୍.
ଭବେ ଶର୍କରା ମାତ୍ର (OGTT)
ହେପାଟାଇଟିସ୍-ବି
ଏଚ୍.ଆଇ.ଭି.
ଡିପ୍ରିସିସ୍

୨ଟି ଟିଟାନସ୍ ଡିପ୍ରେସିଭିଆ (ଟିଟି) ଟାକା ନିଅନ୍ତୁ । ପ୍ରଥମ ଟିଟି ଗର୍ଭାବସ୍ଥାରେ ସେବାଳ ଜାଣିବା ପରେ ଏବଂ ଦ୍ୱିତୀୟ ଟିଟି ଏବଂ ମାସ ପରେ ନିଅନ୍ତୁ ।
ଯଦି ପୂର୍ବ ଗର୍ଭ ୩ ବର୍ଷ ଭିତରେ ହୋଇଥିବା ଏବଂ ଦ୍ୱିତୀୟ ଟିଟାନସ୍ ଡିପ୍ରେସିଭିଆ ଟାକା ନେଇଥାନ୍ତି, ତାହେଲେ ଗୋଟିଏ ଟିଟାନସ୍ ଡିପ୍ରେସିଭିଆ ବୁଝାଇ ଟାକା ନିଅନ୍ତୁ ।

ଗର୍ଭାବସ୍ଥା ପୂର୍ବରୁ ସାପ୍ତାହିକ ଆଇ.ଏଫ୍.ଏ. ବଟିକା ଖାଇଥିଲେ କି ? ☐ ହଁ ☐ ନାଁ

ପ୍ରତ୍ୟେକ ଦିନ ଅତି କମ୍‌ରେ ଗୋଟିଏ ଲେଖାଏଁ କୌଣସି ବଟିକା ଗର୍ଭ ଧାରଣର ବର୍ତ୍ତମାନ ମାସ ଠାରୁ ପ୍ରଥମ ପର୍ଯ୍ୟନ୍ତ ଖାଆନ୍ତୁ । ଗର୍ଭଧାରଣର ୧୪ ସପ୍ତାହ ଠାରୁ ପ୍ରଥମ ପର୍ଯ୍ୟନ୍ତ ପ୍ରତିଦିନ ୨ଟି ଲେଖାଏଁ ଡାଲ୍‌ସିସ୍ ବଟିକା ଖାଆନ୍ତୁ । କୌଣସି ଡାଲ୍‌ସିସ୍ ବଟିକା ବ୍ୟବହାର ନାହିଁ ଖାଆନ୍ତୁ ।

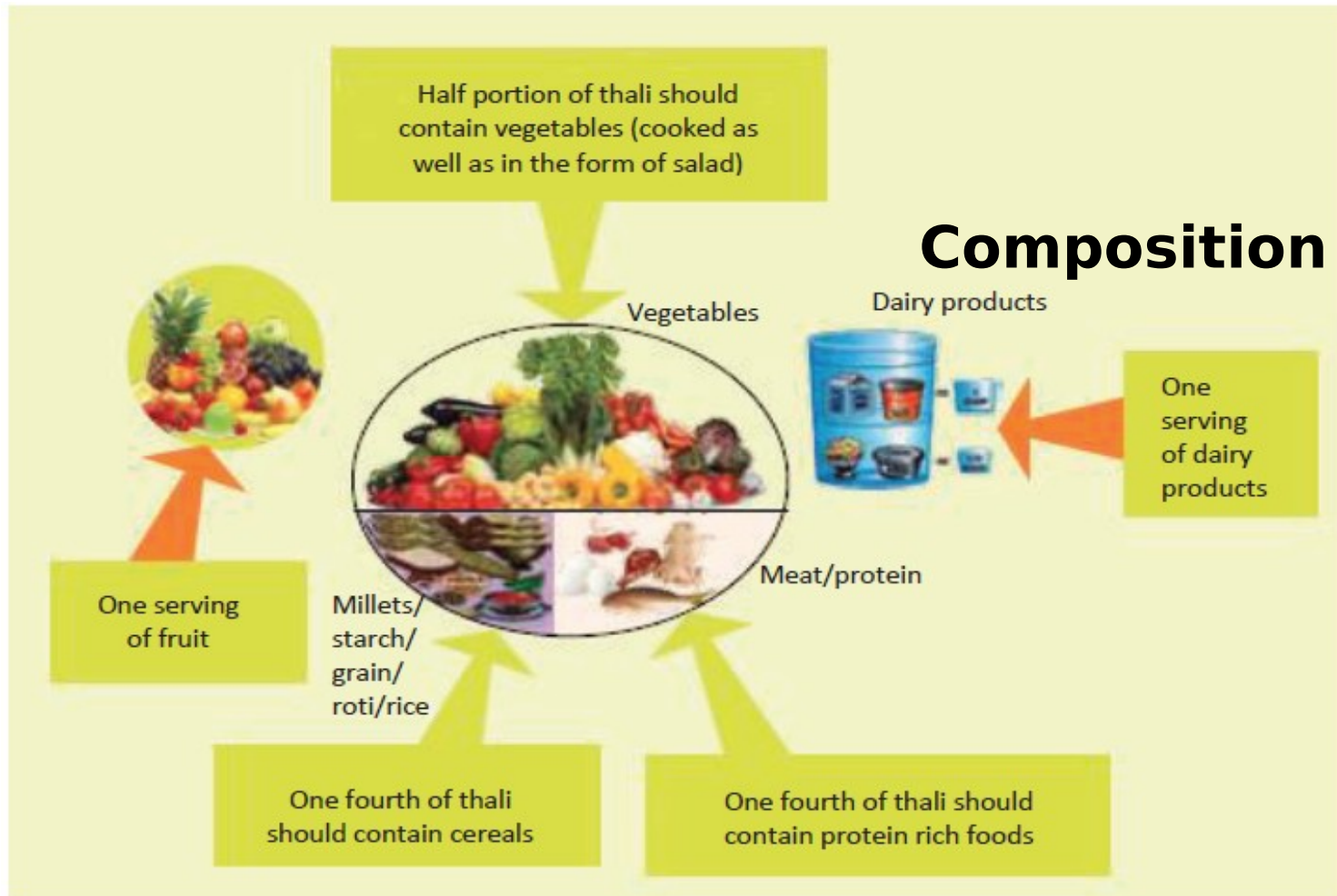
Management of GDM-Medical Nutrition Therapy (MNT)

1. Single visit approach should be used for GDM diagnosis and MNT counseling- all pregnant women with GDM should receive MNT counseling on the day of diagnosis.
2. The meal pattern should provide adequate calories and nutrients to the woman and the baby in the womb to meet the needs of the pregnancy
3. The meal plan should be individualized and such that prevents excess weight gain and post-prandial hyperglycemia

$$\begin{array}{ccccccc} \text{Energy} & & & & & & \text{Energy} \\ \text{Requireme} & & & & & & \text{requirement} \\ \text{nt of} & = & \text{nt of an} & + & \text{required} & + & \text{for increase in} \\ \text{pregnant} & & \text{adult} & & \text{for foetal} & & \text{body weight of} \\ \text{woman} & & \text{woman} & & \text{growth} & & \text{pregnant} \\ & & & & & & \text{woman} \end{array}$$

Medical Nutrition Therapy

Meal plan should be divided in to 3 major meals-breakfast, lunch and dinner and 2-3 snacks in between them



Management of GDM – Physical Exercise (contd.)

Do's	Don'ts
Do moderate and regular physical activity	Do not get too tired while exercising
Wear light and loose comfortable clothes during exercise	Do not do any activity while lying on back/abdomen after beginning of 2 nd trimester
Drink sips of water before, during and after exercise	Do not exercise in hot weather
Stop when you feel tired or have difficulty in talking	Do not perform exercise that may bump or hurt belly. Do not do exercises which need balancing
Eat healthy diet	Do not fast or skip a meal
Choose activities which do not require standing or balancing	Do not over exert

Physical Exercise

